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Escuela Ingeniería Informática

**TRABAJO DE FIN DE GRADO
GRADO EN INGENIERÍA INFORMÁTICA
DEL SOFTWARE**

Evaluación de la influencia de colores en interfaces en
modo oscuro mediante eye-tracking

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Chapter 1

Acknowledgments

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Chapter 2

Abstract

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Chapter 3

Introduction

3.1 Context

Computer scientists have always been seen as “hackers” that work with Command Line Interfaces (CLI). This belief comes from a long-gone era where Graphical User Interfaces (GUI) wouldn’t exist. In reality, Command Line Interfaces and Graphical User Interfaces live together in peace in the different sectors of informatics. However, lately the Graphical User Interfaces are experiencing, perhaps what can be called a ‘golden era’. The evolution from the first Graphical User Interface computer, the Xerox Alto in 1973[9], to Web GUIs applications still share the same core. However, they have quickly evolved in the past years thanks to eased access to the Internet and computers. Nowadays, big companies dictate the cutting edge of technology regarding GUIs. A trend that has become obvious is the rise of Dark mode, dividing the whole user community of many applications.

3.2 Motivation

Year after year we have seen how the whole world has changed at a really high pace. Two of the fastest changing areas are Graphical User Interfaces and User Experience (UX). A long time has passed since the times when developers would consider the front end a secondary part of development.

Nowadays a new problem has become viral, not only inside the informatics world but throughout the whole spectrum of internet users: is dark mode better (from now on negative polarity) or bright mode (from now on positive polarity)? This has led many companies to embrace dark mode as its corporate colors, for example the social network X (former Twitter) shifted to negative polarity as its default mode.

There have been many studies regarding usability between both options. However, no one has actually studied the effects of the positive and negative display polarity in longer texts. That has risen the following questions:

- Does the user get more fatigued while reading in negative polarity?
- Will they read faster or slower?

- Will the user struggle to find more information?
- E-commerce websites are mainly programmed in positive polarity. Is the negative polarity really a disadvantage for the user experience while reading and analyzing a product?

All these questions will be analyzed and given the best answer possible in this thesis. A conclusion regarding long texts and display polarities will be studied and explained here.

3.3 Purpose of the thesis

The purpose of the thesis is to investigate the visual fatigue and readability of both display polarities in terms of reading long texts in web-based products. This will be achieved by using a tobii eye-tracker. This tool can give real time data of the eye, giving us objective and solid information.

Visual fatigue can be defined as a moment in which the eyes have a huge workload, and the user can perceive a sensation of discomfort. Visual fatigue can happen after a long period of time in front of a screen, but it also happens every time someone reads a text on a smaller scale. It is usually studied with questionnaires leading to subjectiveness in the analysis. However, in recent times, tools such as eye-trackers have proven a new way to study it. Symptoms of visual fatigue can be calculated as the time the user takes to read a whole paragraph and how many times the person takes to stop at each word.

3.4 Document Structure

The structure of the document has been briefly inspired by Jose Manuel Redondo's spreadsheet of final thesis for development thesis [4]. However, in order to adapt it to an investigation thesis the spreadsheet has been adapted and now doesn't follow non specific standard. This new "version" is mostly inspired by the scientific method by expanding the common topics on a scientific paper, while at the same time being mixed with the most common sections of a usual informatics thesis.

Chapter 4

State of the Art

The state of the art is the information relevant to the study that will be done. In this way, we can close the scope of what this new investigation will prove and find studies that can help us close that new research gap. As a small reminder, display polarity is the mode that will be shown on the screen, positive polarity being dark characters in bright background and negative polarity the other way around.

4.1 Previous Studies

The previous studies section, as explained above, consists of doing research in order to slowly close the research gap. This process was done by following a “baking” process, where an initial research was done on the subject in several Databases such as IEEExplore[1] or Scopus[7], refining the keywords every time to find related articles. The best keywords appeared to be:

- Display polarity
- Visual fatigue
- Reading

After reading and analyzing the first bake, a second bake of refinement and vertical expansion (analyzing cites and references of such scientific papers) was done. After erasing the unrelated articles and the ones out of scope, the following key articles were found:

4.1.1 The impact of web page text-background colour combinations on readability, retention, aesthetics and behavioural intention[3]

Table 4.1: Summary of the article by Hall and Hanna

Authors	Hall, Richard H Hanna, Patrick
Year	2004
Source	Behaviour and Information Technology
DOI	10.1080/01449290410001669932

Reason for choice

The motivation to use this paper as a reference is because of the Likert scale questionnaire that is used, giving questions that can perfectly fit with the desired study. In addition, they hand out more questions that could be applied to further explain other non-studied parts of my thesis.

Abstract

The purpose of this experiment was to examine the effect of web page text/background colour combination on readability, retention, aesthetics, and behavioural intention. One hundred and thirty-six participants studied two Web pages, one with educational content and one with commercial content, in one of four colour-combination conditions. Major findings were: (a) Colours with greater contrast ratio generally lead to greater readability; (b) colour combination did not significantly affect retention; (c) preferred colours (i.e., blues and chromatic colours) led to higher ratings of aesthetic quality and intention to purchase; and (d) ratings of aesthetic quality were significantly related to intention to purchase.

4.1.2 Study on the effects of display color mode and luminance contrast on visual fatigue. [10]

Table 4.2: Summary of the article by Xie *et al.*

Authors	Xie, Xiaojiao Song, Fanghao Liu, Yan Wang, Shurui Yu, Dong
Year	2021
Source	IEE Access
DOI	10.1109/ACCESS.2021.3061770

Reason for choice

This study is one of the closest to what the purpose of this thesis is related to. However, there are some differences. For example, the aim is different, and they are not focused on usability. Nevertheless, this paper has provided good information to reach the desired experiment for investigating the usability and visual fatigue. In the following illustration we can see the luminance contrast: (a) 0.969, (b) 0.935, (c) 0.855, (d) 0.868, (e) 0.725, (f) 0.469



Figure 4.1: Experimental Illustrations.

Abstract

Using electronic devices at night can easily cause visual fatigue. We investigated the conjoint effects of color mode and luminance contrast on visual fatigue and subjective preference when using electronic devices under low screen luminance and low ambient illumination at night. A multidimensional approach based on eye and subjective measures was used to test 2 color modes (dark mode, light mode) and 6 luminance contrast ratios (0.969, 0.935, 0.868, 0.855, 0.725, 0.469) in a 2×6 experimental design. We used eye movement tracking technology to collect blink rate and pupil diameters, and used the Likert scale to measure subjective visual fatigue scale and preference.

Results showed that reading in the dark mode reduced visual fatigue, as reflected by an increase in blink rate and pupil accommodation. Lower subjective visual fatigue scale and higher preference were found in the light mode due to subjects' using habits of dark texts on a light background. There was a significant negative correlation between (text-background) luminance contrast and visual fatigue, and subjects preferred higher luminance contrast. We observed the lowest visual fatigue under the luminance contrast of 0.969 in the dark mode, and the lowest subjective preference when the luminance contrast was lower than 0.725. We suggest the users should choose the dark mode to reduce visual fatigue when using electronic devices at night. These findings also provide a reference for the design of interactive interfaces such as tablets and mobile phones, and have practical implications for reducing visual fatigue.

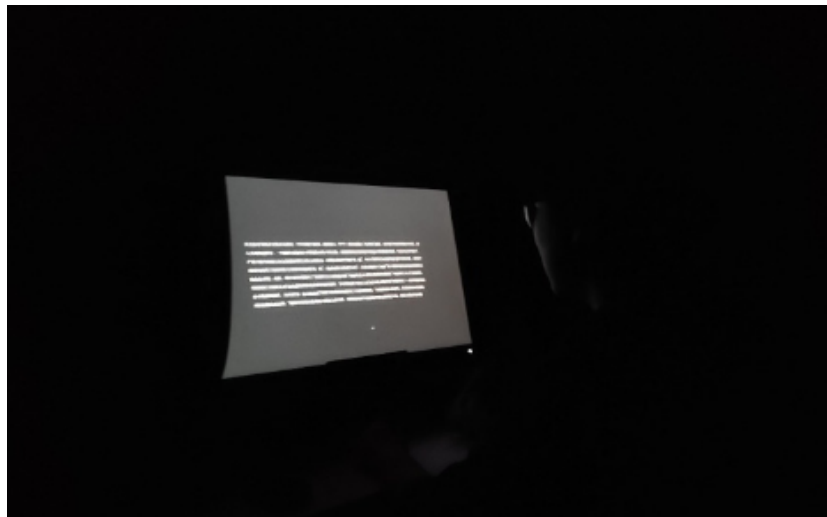


Figure 4.2: Photo of participant during the experiment

4.1.3 Web Design Attributes in Building User Trust, Satisfaction, and Loyalty for a High Uncertainty Avoidance Culture.[2]

Table 4.3: Summary of the article by Faisal *et al.*

Authors	Faisal, C M Gonzalez-Rodríguez, Martin Fernandez-Lanvinv, Daniel De Andres-Suarez, Javier
Year	2016
Source	IEEE Access
DOI	10.1109/THMS.2016.2620901

Reason for choice

In order to extract usability subjective data, it is necessary to use a Likert questionnaire. The display polarity questions are already selected. However, there are a few general preference questions that need to be asked. It is better to obtain well studied questions rather than using self-created ones.

Abstract

In this study, we attempt to evaluate the user preferences for web design attributes (i.e., typography, color, content quality, interactivity, and navigation) to determine the trust, satisfaction, and loyalty for uncertainty avoidance cultures. Content quality and navigation have been observed as strong factors in building user trust with e-commerce websites. In contrast, interactivity, color, and typography have been observed as strong determinants of user satisfaction. The most relevant and interesting finding is related to typography, which has been rarely discussed in e-commerce literature. A questionnaire was designed to collect data to corroborate the proposed model and hypotheses. Furthermore, the partial least-squares method was adopted to analyze the collected data from the students who participated in the test ($n = 558$). Finally, the results of this study provide strong support to the proposed model and hypotheses. Therefore, all the web design attributes were observed as important design features to develop user trust and satisfaction for uncertainty avoidance cultures. Although both factors seem to be relevant, the relationship between trust and loyalty was observed to be stronger than between satisfaction and

loyalty; thus, trust seems to be a stronger determinant of loyalty for risk/high uncertainty avoidance cultures.

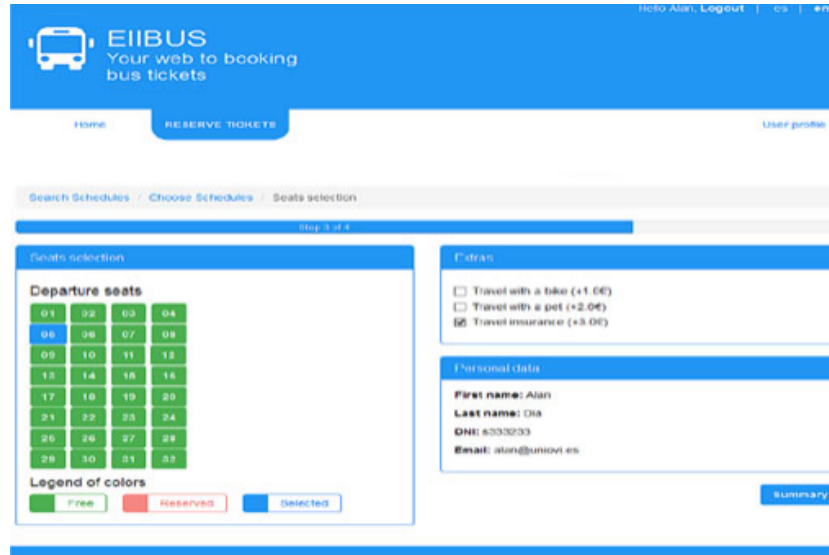


Figure 4.3: E-ticket bus website prototype used in the experiment.

4.1.4 **The Effect of Display Polarity on Reading Speed
and Reading Error Among Young Adults.[5]**

Table 4.4: Summary of the article by Muhamad and Mokhtar

Authors	Muhamad, Nurulain Mokhtar, Nurliyana
Year	2024
Source	Journal of Health Science and Medical Research
DOI	10.31584/jhsmr.20241095

Reason for choice

The reason for choosing this article is the fact that, even though the study is quite similar, they are not using an eye tracker but instead measuring times by hand, hence giving a general measurement with a lot of imprecisions. However, the methodology (without considering the tools for measuring) is an acceptable one, and some of it is considered while processing the data.

Abstract

Objective: The widespread adoption of digital devices has surged, particularly since the coronavirus disease 2019 (COVID-19) pandemic. Nearly everyone now owns devices like laptops, tablets, or smartphones, offering options for light mode (positive polarity) and dark mode (negative polarity) to suit individual preferences. This study examines how display polarity affects reading performance among young adults. Material and Methods: Thirty participants engaged in a 15-minute reading task on a laptop with randomly assigned display polarities, followed by a 15-minute break before repeating the task. Results: Reading speed, measured in words per minute (wpm), differed significantly between polarities, with negative polarity yielding higher speeds (136.27 ± 25.58 wpm) compared to positive polarity (128.42 ± 19.98 wpm), $Z = -2.355$, $p\text{-value} < 0.05$. However, no significant polarity-related differences were found in reading errors, including mispronunciation ($p\text{-value} = 0.193$) or omission ($p\text{-value} = 0.113$). Conclusion: Negative polarity displays enhanced reading performance by increasing reading speed; while reading errors remained unaffected.

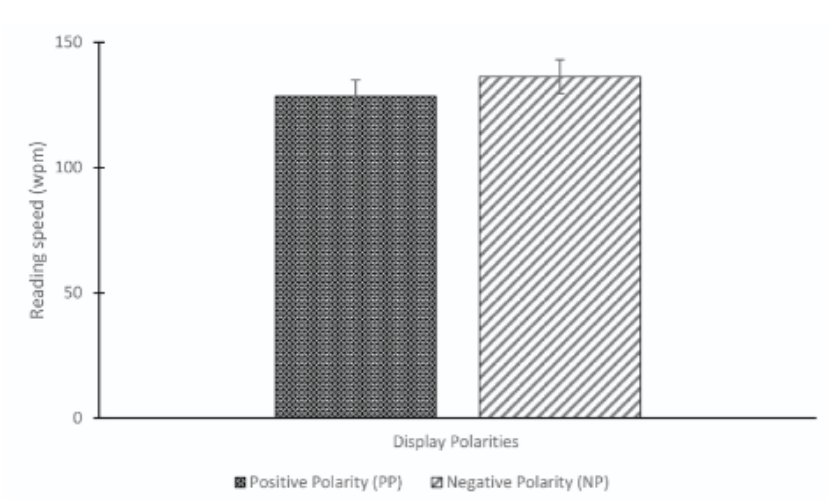


Figure 4.4: The comparison of reading speed between positive and negative display polarities.

4.1.5 Eye Tracking in Human-Computer Interaction and Usability Research: Current Status and Future Prospects [6]

Table 4.5: Summary of the article by Poole and Ball

Authors	Poole, Alex Ball, Linden J.
Year	2006
Source	Encyclopedia of Human Computer Interaction
DOI	10.4018/978-1-59140-562-7.ch034

Reason for choice

BLABLABLA TODO

Abstract

Eye-movement tracking is a method that is increasingly being employed to study usability issues in HCI contexts. The objectives of the present chapter are threefold. First, we introduce the reader to the basics of eye-movement technology, and also present key aspects of practical guidance to those who might be interested in using eye tracking in HCI research, whether

in usability-evaluation studies, or for capturing people's eye movements as an input mechanism to drive system interaction. Second, we examine various ways in which eye movements can be systematically measured to examine interface usability. We illustrate the advantages of a range of different eye-movement metrics with reference to state-of-the-art usability research. Third, we discuss the various opportunities for eye-movement studies in future HCI research, and detail some of the challenges that need to be overcome to enable effective application of the technique in studying the complexities of advanced interactive-system use.

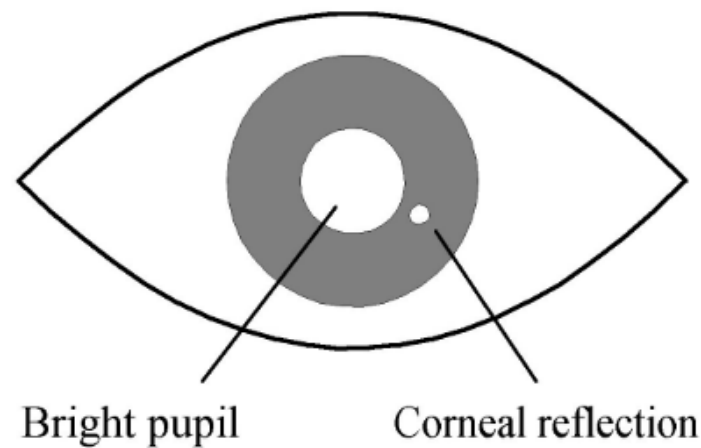


Figure 4.5: Corneal reflection and bright pupil as seen in the infrared camera such as Tobii Pro Nano [8].

4.2 Research Gap

There aren't many investigations regarding display polarity. However, the few existing ones are inconsistent. Manual measuring methods, such as taking time with a chrono, are usual procedures. Many times, the total populations could reach less than 20 people in total. Therefore, the research gap remains broad. In the case of this thesis, the investigation will be conducted for web-based wikis and product descriptions, as this can help to study the long text. It is also important to note that the aim is to give consistent results that are backed up by statistical methods and precise tools such as the Tobii eye-tracker.

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